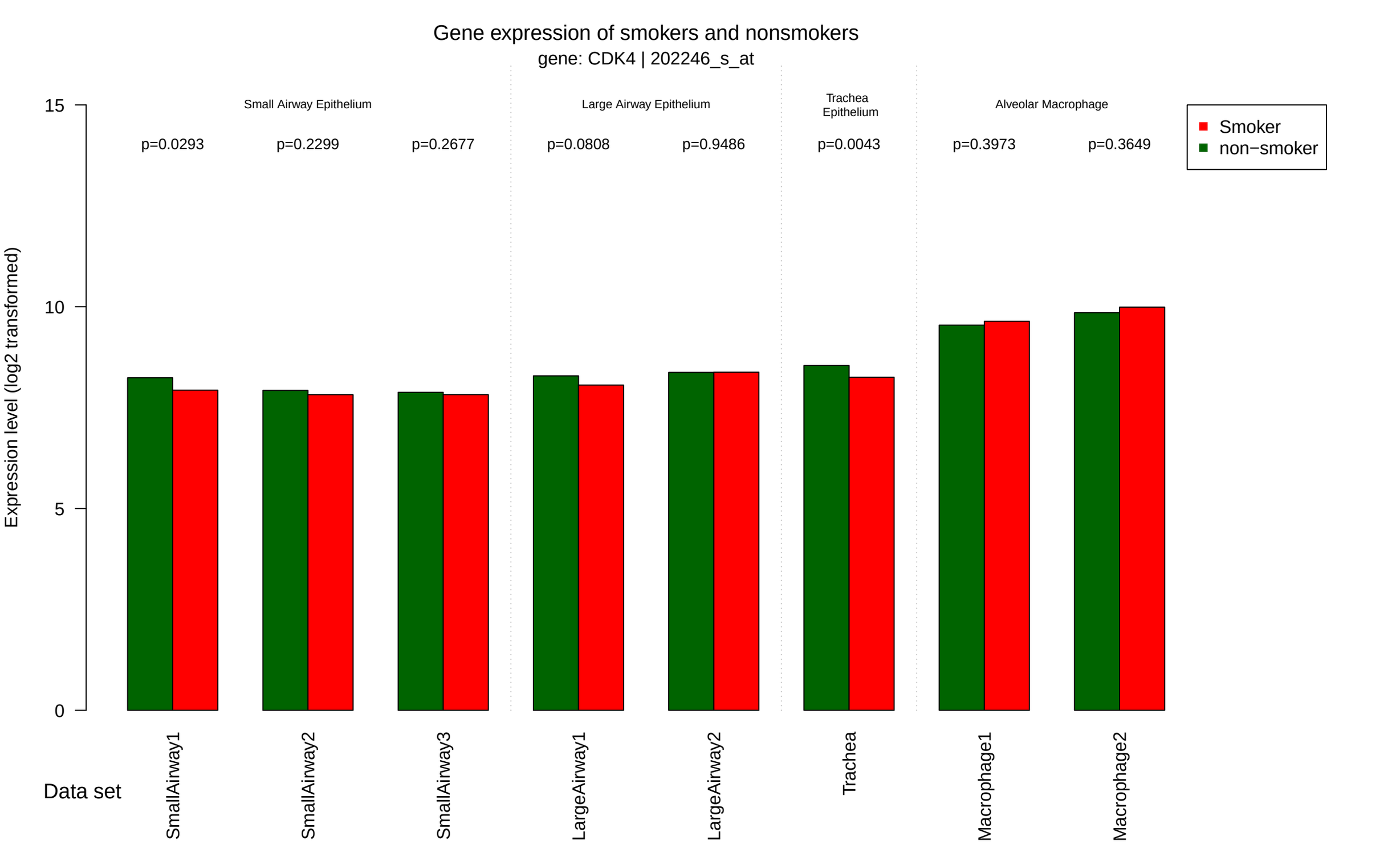
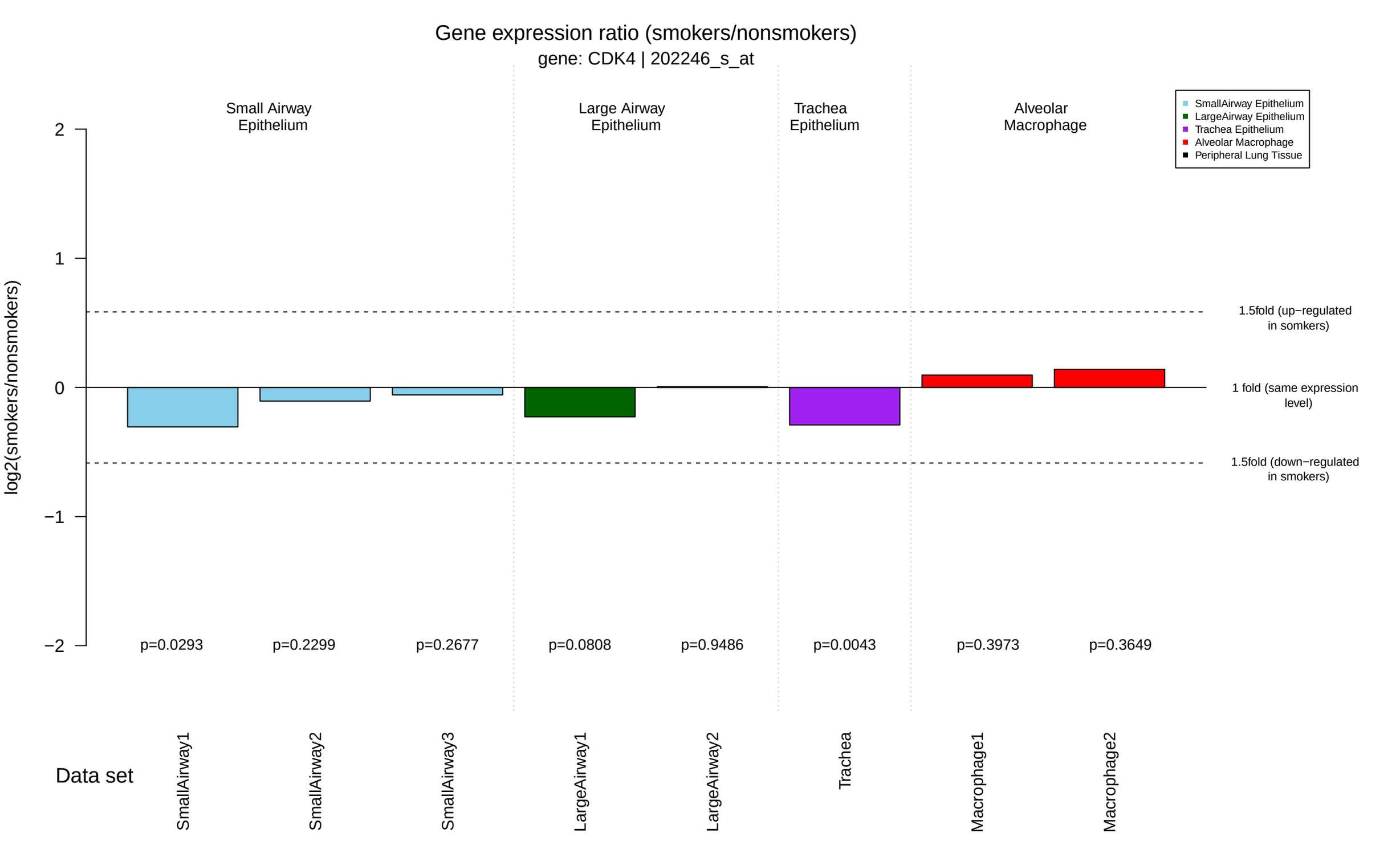
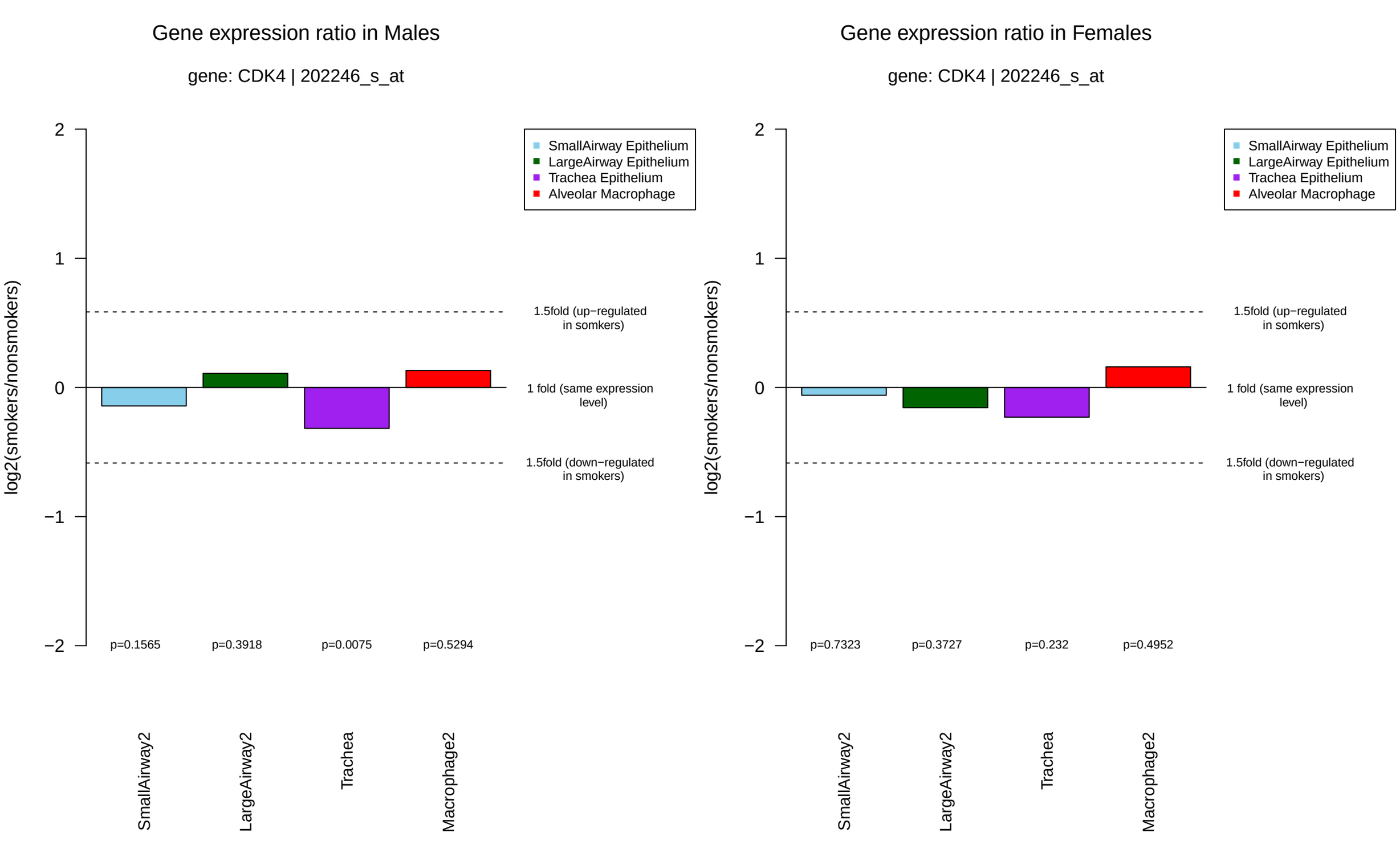
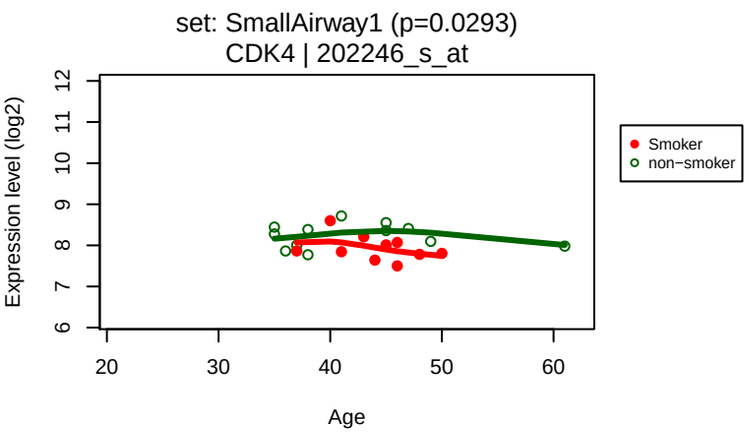
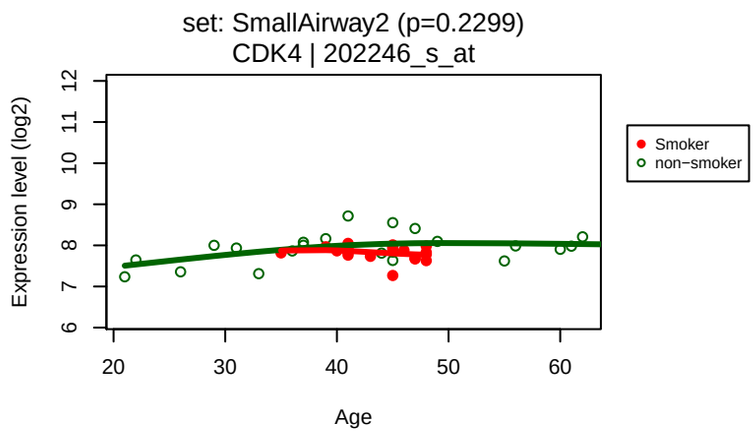
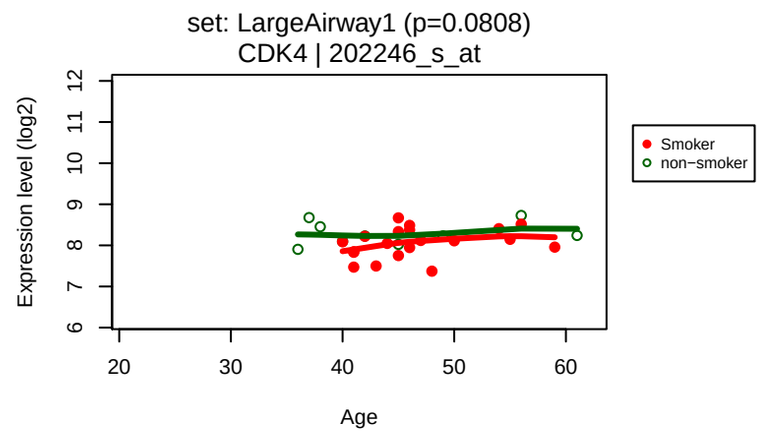

Gene expression of smokers and nonsmokers
gene: CDK4 | 202246_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0293
p=0.2299
p=0.2677
p=0.0808
p=0.9486
p=0.0043
p=0.3973
p=0.3649
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

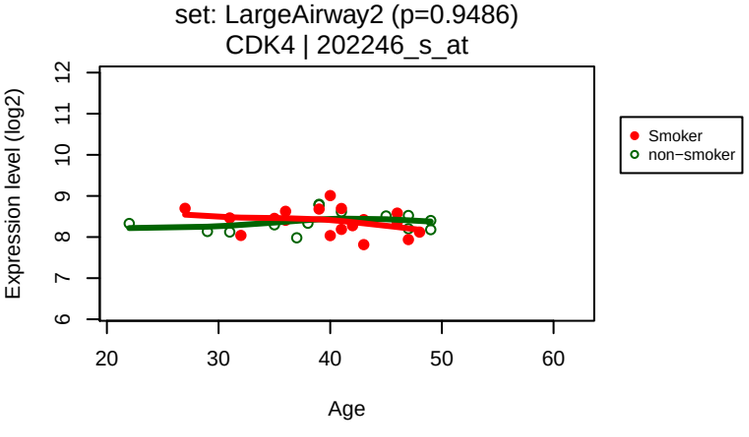

Gene expression ratio (smokers/nonsmokers)
gene: CDK4 | 202246_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0293
p=0.2299
p=0.2677
p=0.0808
p=0.9486
p=0.0043
p=0.3973
p=0.3649
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

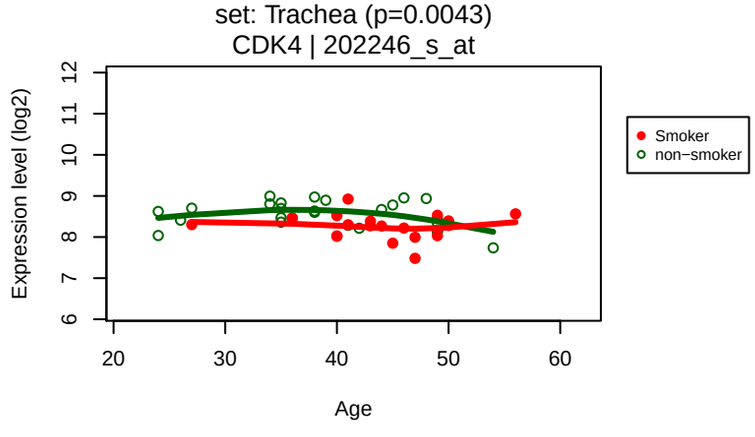

Gene expression ratio in Males
gene: CDK4 | 202246_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1565
p=0.3918
p=0.0075
p=0.5294
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CDK4 | 202246_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7323
p=0.3727
p=0.232
p=0.4952
SmallAirway2
LargeAirway2
Trachea
Macrophage2

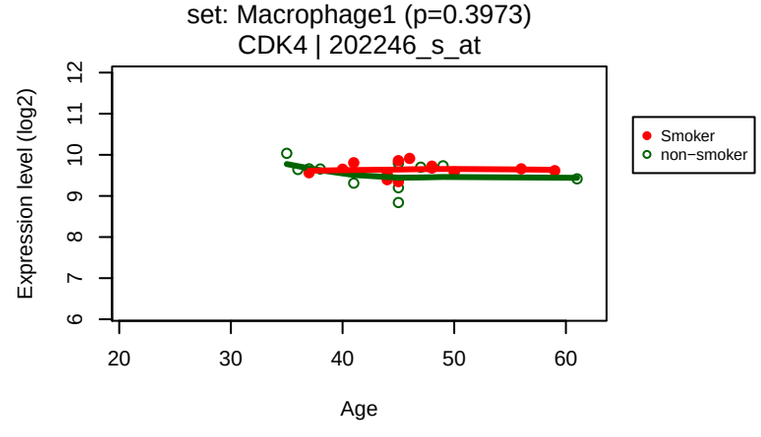

set: SmallAirway1 (p=0.0293)
CDK4 | 202246_s_at
Expression level (log2)
Age
Smoker
non−smoker

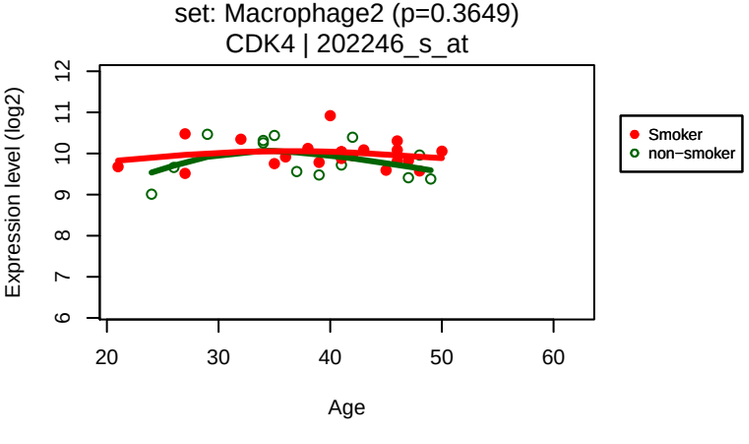

set: SmallAirway2 (p=0.2299)
CDK4 | 202246_s_at
Expression level (log2)
Age
Smoker
non−smoker

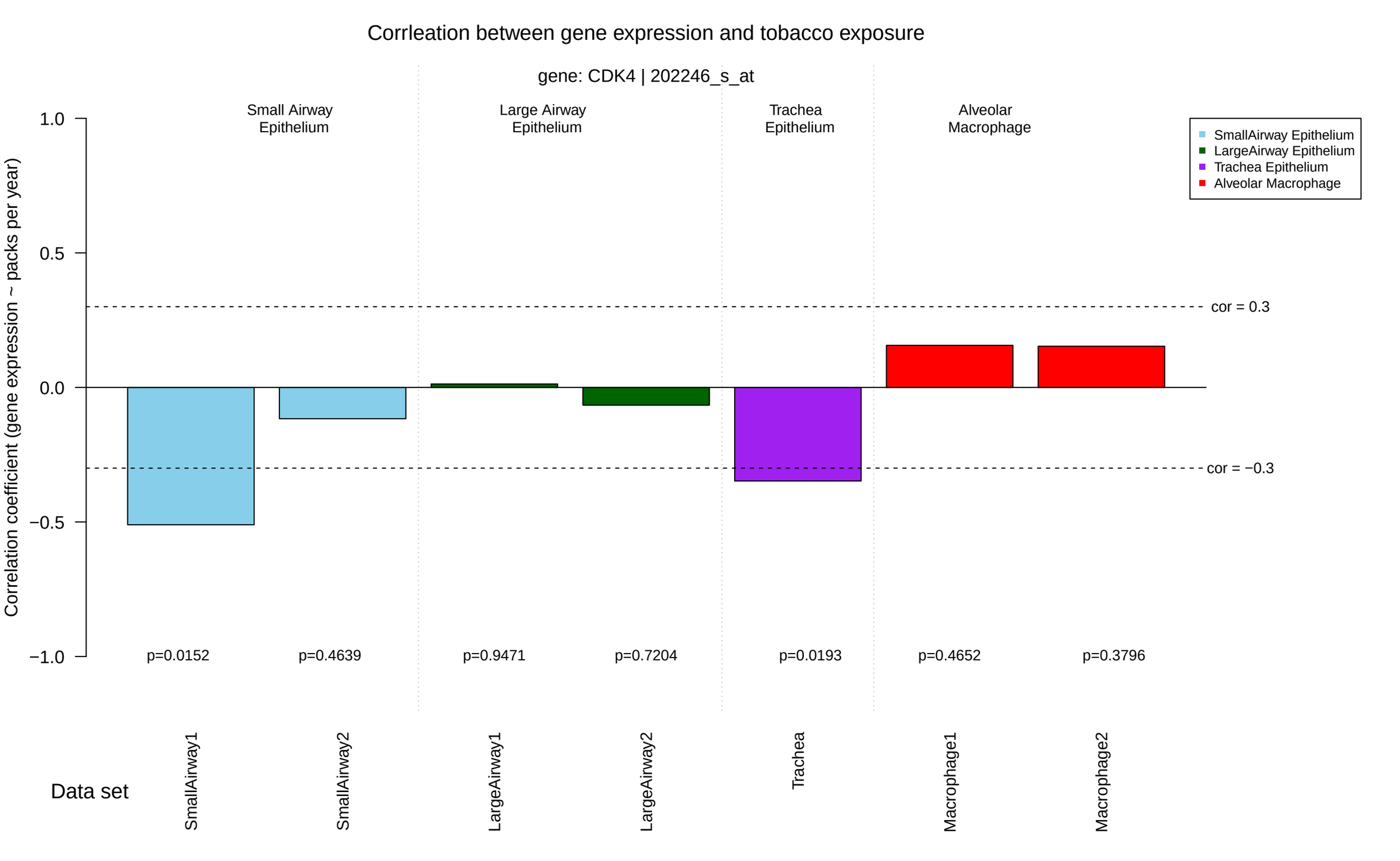

set: LargeAirway1 (p=0.0808)
CDK4 | 202246_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9486)
CDK4 | 202246_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0043)
CDK4 | 202246_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3973)
CDK4 | 202246_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3649)
CDK4 | 202246_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CDK4 | 202246_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0152
p=0.4639
p=0.9471
p=0.7204
p=0.0193
p=0.4652
p=0.3796
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2